

Prospective Health Across Contexts

Contextual Licensing, Developmental Transport, and Open-System Support

Zed James

Paper II in a series on the formal and measurement foundations of health.

© 2026 Zed James. This work is licensed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

Preprint – October 4, 2026.
DOI: [10.5281/zenodo.23146281](https://doi.org/10.5281/zenodo.23146281).

Abstract

Prospective health is evaluated by

$$\text{Health}_{\mathfrak{S},r}(x; d, t) \iff \text{Realizes}_{\mathfrak{S}}(x) \wedge \text{Adequate}_{\mathfrak{S},r}(\text{Capacity}_{\mathfrak{S}}(x; d, t)).$$

Paper II studies transport of the resulting capacity representation across context, developmental stage, and open-system support.

For $c = (\mathfrak{S}, d, t)$, whole-query licensing defines

$$L_c = \{r \mid \text{Licensed}(c, r)\}, \quad a \sim_c b \iff \forall r \in L_c, \text{Adequate}_{c,r}(a) \leftrightarrow \text{Adequate}_{c,r}(b),$$

and hence

$$\text{HealthCap}_c = \text{Cap}_c / \sim_c.$$

For a state relation $R : X_0 \rightsquigarrow X_1$ and capacity functions

$$C_0 : X_0 \rightarrow \mathcal{A}, \quad C_1 : X_1 \rightarrow \mathcal{B},$$

capacity determinacy is equivalent to a unique relation-attained factorization

$$\text{CapacityDeterminate}(R, C_0, C_1) \iff \exists! \kappa_R : \mathcal{A}_R \rightarrow \mathcal{B},$$

where $\mathcal{A}_R$ is the set of source capacities attained along R . Applying the same construction to

$$Q_c(x) = [\text{Capacity}_c(x)]_c$$

gives canonical attained transport of health-visible information, an explicit relation-valued image, and its functionality criterion.

A finite developmental fixture separates stage semantics from horizon-dependent capacity. The stage law

$$\{(N, A), (A, E)\}$$

acts on licensing, and a complete two-atom capacity census determines a unique now-to-long capacity map equal to the closed-form recovery operation. Explicit controls exhibit stage-induced quotient refinement, branching and coalescence, raw-capacity indeterminacy with health-visible determinacy, support-dependent factorization obstruction, and the realization and adequacy conditions required for full-health comparison.

The contribution is a health-specific organization of contextual licensing, observational equivalence, relation-generated capacity evolution, factorization, relation-valued transport, and full-health semantics. Lean certifies the displayed definitions, theorems, witnesses, and obstruction calculations.

1 Introduction: the information hierarchy of prospective health

The starting proposition is

$$\boxed{\text{Health}_{\mathfrak{S},r}(x; d, t) \iff \text{Realizes}_{\mathfrak{S}}(x) \wedge \text{Adequate}_{\mathfrak{S},r}(\text{Capacity}_{\mathfrak{S}}(x; d, t))}. \quad (1)$$

The prospective capacity is derived from complete viable histories:

$$\mathcal{V}_{d,t}(x) = \{(y, \gamma) \mid \gamma \in \text{Hist}(d, t, x, y), \text{Lawful}(\gamma), \text{Continues}(\gamma)\}, \quad (2)$$

followed by the declared observer and collector,

$$\text{Capacity}_{\mathfrak{S}}(x; d, t) = \text{collect}(\{o(v) \mid v \in \mathcal{V}_{d,t}(x)\}). \quad (3)$$

Equations (2)–(3) are the fixed-query response and capacity semantics established in Paper I [1]. Paper II adopts these objects unchanged as inherited inputs. In particular, the complete viable-history family is upstream of the declared capacity representation, which may compress distinct viable histories.

For an ordinary context

$$c = (\mathfrak{S}, d, t), \quad q = (c, r), \quad (4)$$

define

$$L_c = \{r \mid \text{Licensed}(c, r)\}, \quad (5)$$

and

$$a \sim_c b \iff \forall r \in L_c, \text{Adequate}_{c,r}(a) \leftrightarrow \text{Adequate}_{c,r}(b). \quad (6)$$

The corresponding health-visible quotient is

$$\boxed{\text{HealthCap}_c = \text{Cap}_c / \sim_c, \quad \pi_c(a) = [a]_c}. \quad (7)$$

For $r \in L_c$,

$$\overline{\text{Adequate}_{c,r}}([a]_c) \iff \text{Adequate}_{c,r}(a), \quad (8)$$

and

$$\text{Health}_{c,r}(x) \iff \text{Realizes}_c(x) \wedge \overline{\text{Adequate}_{c,r}}(Q_c(x)), \quad Q_c(x) := \pi_c(\text{Capacity}_c(x)). \quad (9)$$

1

The paper uses the hierarchy

$$\boxed{\begin{array}{ccccc} x & \xrightarrow{\text{Capacity}_c} & a & \xrightarrow{\pi_c} & [a]_c = Q_c(x) \\ \text{state} & & \text{raw capacity} & & \text{health-visible information.} \end{array}} \quad (10)$$

A contextual relation $x R y$ may induce a deterministic raw-capacity map, a deterministic health-visible map, a relation-valued health-class image, or only a full-health comparison after further realization and adequacy hypotheses. The sections below keep these layers separate.

Whole-query licensing remains primitive. The broader governance record also contains declaration, empirical-support, and provenance fields; the transport theorems in Paper II use the licensing coordinate. A common licensed language over a fixed-specification context family exists exactly when licensing is context-invariant over that family. ²

¹Formal counterpart: `lean/Tier33/Health/Paper/FAQ/ContextualLicensedQuotient.lean::contextLicensedAdequacy_descends;contextLicensedHealth_descends.`

²Formal counterpart: `lean/Tier33/Health/Paper/FAQ/ContextualLicensing.lean::commonLicensedLanguage_exists_iff.`

2 Quotient descent for a supplied capacity map

Let

$$\kappa : \text{Cap}_{c_0} \rightarrow \text{Cap}_{c_1}.$$

Define quotient compatibility by

$$\boxed{\text{QuotCompat}(\kappa) : \iff \forall a, b, a \sim_{c_0} b \implies \kappa(a) \sim_{c_1} \kappa(b).} \quad (11)$$

Theorem 2.1 (Contextual quotient descent). *Condition (11) holds if and only if there exists a unique*

$$\bar{\kappa} : \text{HealthCap}_{c_0} \rightarrow \text{HealthCap}_{c_1}$$

such that

$$\boxed{\bar{\kappa}([a]_{c_0}) = [\kappa(a)]_{c_1} \quad \forall a.} \quad (12)$$

Equivalently, the square

$$\begin{array}{ccc} \text{Cap}_{c_0} & \xrightarrow{\kappa} & \text{Cap}_{c_1} \\ \pi_{c_0} \downarrow & & \downarrow \pi_{c_1} \\ \text{HealthCap}_{c_0} & \xrightarrow{\bar{\kappa}} & \text{HealthCap}_{c_1} \end{array}$$

commutes.

Proof. Define

$$\bar{\kappa}([a]_{c_0}) := [\kappa(a)]_{c_1}.$$

If $[a]_{c_0} = [b]_{c_0}$, then $a \sim_{c_0} b$, so quotient compatibility gives

$$\kappa(a) \sim_{c_1} \kappa(b),$$

hence

$$[\kappa(a)]_{c_1} = [\kappa(b)]_{c_1}.$$

Thus the definition is independent of representative. Equation (12) fixes the value of any commuting map on every quotient class, proving uniqueness.

Conversely, if a commuting $\bar{\kappa}$ exists and $a \sim_{c_0} b$, then

$$[a]_{c_0} = [b]_{c_0},$$

and therefore

$$[\kappa(a)]_{c_1} = \bar{\kappa}([a]_{c_0}) = \bar{\kappa}([b]_{c_0}) = [\kappa(b)]_{c_1}.$$

Hence $\kappa(a) \sim_{c_1} \kappa(b)$. □

3

Negating the compatibility condition gives the exact obstruction:

$$\boxed{\neg \exists \bar{\kappa} \text{ satisfying Equation (12)} \iff \exists a, b, a \sim_{c_0} b \wedge \kappa(a) \not\sim_{c_1} \kappa(b).} \quad (13)$$

³Formal counterpart:

`lean/Tier33/Health/Paper/FAQ/ContextTransport.lean::contextQuotientTransport_existsUnique_iff.`

A commuting function assigns one target class to each source class, so the displayed split is the exact obstruction.⁴

This is standard quotient mathematics. The health-specific structure enters through the contextual equivalence relations $\sim_c$.

3 Relation-attained factorization

A developmental or environmental model may provide a relation between source and target states before it provides any capacity map. Let

$$R : X_0 \rightsquigarrow X_1, \quad C_0 : X_0 \rightarrow \mathcal{A}, \quad C_1 : X_1 \rightarrow \mathcal{B}. \quad (14)$$

Define the relation-attained source carrier

$$\boxed{\mathcal{A}_R = \{a \in \mathcal{A} \mid \exists x, y, R(x, y) \wedge C_0(x) = a\}.} \quad (15)$$

Definition 3.1 (Capacity determinacy).

$$\boxed{\text{CapacityDeterminate}(R, C_0, C_1) : \iff \forall x, x', y, y', \\ R(x, y) \wedge R(x', y') \wedge C_0(x) = C_0(x') \\ \implies C_1(y) = C_1(y').} \quad (16)$$

Theorem 3.2 (Attained factorization). *Capacity determinacy holds if and only if there exists a unique*

$$\kappa_R : \mathcal{A}_R \rightarrow \mathcal{B}$$

satisfying

$$\boxed{\kappa_R(C_0(x)) = C_1(y) \quad \text{for every } R(x, y).} \quad (17)$$

Proof. Take $a \in \mathcal{A}_R$. Choose x, y with

$$R(x, y), \quad C_0(x) = a,$$

and set

$$\boxed{\kappa_R(a) := C_1(y).} \quad (18)$$

For another choice x', y' ,

$$R(x', y'), \quad C_0(x') = a = C_0(x),$$

capacity determinacy gives

$$C_1(y') = C_1(y),$$

so Equation (18) is well-defined.

If $f : \mathcal{A}_R \rightarrow \mathcal{B}$ also commutes with every related pair, choose x, y attaining a . Then

$$f(a) = f(C_0(x)) = C_1(y) = \kappa_R(a),$$

⁴Formal counterpart: `lean/Tier33/Health/Paper/FAQ/ContextTransportObstruction.lean::contextQuotientTransport_obstruction_iff_splitFiber.`

so $f = \kappa_R$.

Conversely, suppose a commuting κ_R exists. For related pairs with

$$C_0(x) = C_0(x'),$$

the associated attained source values are equal, and

$$C_1(y) = \kappa_R(C_0(x)) = \kappa_R(C_0(x')) = C_1(y').$$

□

5

The exact obstruction is

$$\boxed{\neg \exists \kappa_R : \mathcal{A}_R \rightarrow \mathcal{B} \text{ satisfying Equation (17)} \\ \iff \exists x, x', y, y', R(x, y), R(x', y'), C_0(x) = C_0(x'), C_1(y) \neq C_1(y').} \quad (19)$$

Indeed a deterministic map would have to assign the same output to the common source observation and simultaneously equal two distinct target observations.⁶

Attained uniqueness extends to the entire source carrier when every source-capacity value is relation-attained:

$$\boxed{\mathcal{A}_R = \mathcal{A} \implies \text{attained uniqueness extends to global uniqueness.}} \quad (20)$$

Indeed, for $a \in \mathcal{A} = \mathcal{A}_R$, choose $x \in X_0$ and $y \in X_1$ with

$$R(x, y), \quad C_0(x) = a.$$

If $f, g : \mathcal{A} \rightarrow \mathcal{B}$ both commute with every related pair, then

$$\boxed{f(a) = f(C_0(x)) = C_1(y) = g(C_0(x)) = g(a).}$$

The hypothesis $\mathcal{A}_R = \mathcal{A}$ adds relation-attained coverage to ordinary surjectivity of C_0 : every source value is attained by a source state participating in an R -transition. In the later graph-identity finite fixture, every state participates and ordinary source surjectivity therefore implies Equation (20).

4 Relation-valued health-visible transport

Let

$$Q_i(x) = [\text{Capacity}_{c_i}(x)]_{c_i}, \quad i = 0, 1.$$

⁵Formal counterpart:

`lean/Tier30/Research/RelationalAdmissibility/Realization/AttainedObservationTransport.lean::attainedObservationTransport_existsUnique_iff.`

⁶Formal counterpart:

`lean/Tier33/Health/Transport/AttainedCapacity.lean::attainedCapacityTransport_obstruction_iff.`

The relation-valued health-visible image is defined explicitly by

$$u \bar{R} v \iff \exists x \in X_0, \exists y \in X_1, R(x, y) \wedge Q_0(x) = u \wedge Q_1(y) = v. \quad (21)$$

Its attained source is

$$U_R = \mathbf{Attained}_R(Q_0) = \{u \in \mathbf{HealthCap}_{c_0} \mid \exists x, y, R(x, y) \wedge Q_0(x) = u\}. \quad (22)$$

For every $u \in U_R$, the witnessing pair supplies a target class:

$$u \in U_R \implies \exists v, u \bar{R} v. \quad (23)$$

7

Define health-class determinacy by

$$\mathbf{HealthClassDeterminate}(R; Q_0, Q_1) : \iff R(x, y), R(x', y'), \\ Q_0(x) = Q_0(x') \implies Q_1(y) = Q_1(y'). \quad (24)$$

Theorem 4.1 (Functionality criterion). *The attained relation $\bar{R}|_{U_R}$ is functional exactly when $\mathbf{HealthClassDeterminate}(R; Q_0, Q_1)$ holds.*

Proof. If $u \bar{R} v_1$ and $u \bar{R} v_2$, choose witnesses

$$R(x, y_1), \quad R(x', y_2),$$

with

$$Q_0(x) = u = Q_0(x'), \quad Q_1(y_i) = v_i.$$

Health-class determinacy gives

$$v_1 = Q_1(y_1) = Q_1(y_2) = v_2.$$

Conversely, take two related pairs with equal source classes. They produce two $\bar{R}$ -edges from the same attained source class; single-valuedness forces equality of the target classes. $\square$

8

The corresponding deterministic factorization is

$$\mathbf{HealthClassDeterminate}(R; Q_0, Q_1) \iff \exists! T_R : U_R \rightarrow \mathbf{HealthCap}_{c_1} \quad (25)$$

with $T_R(Q_0(x)) = Q_1(y)$ for every $R(x, y)$.⁹

Representative-level reflection uses a distinct saturation condition:

$$Q_0(x) = Q_0(x'), \quad Q_1(y) = Q_1(y') \\ \implies (R(x, y) \leftrightarrow R(x', y')). \quad (26)$$

⁷Formal counterpart:

`lean/Tier33/Health/Transport/RelationalHealth.lean::attainedRelationalHealthTransport_leftTotal.`

⁸Formal counterpart: `lean/Tier33/Health/Transport/RelationalHealth.lean::attainedRelationalHealthTransport_singleValued_iff;relationalHealthTransport_functional_iff_determinate.`

⁹Formal counterpart:

`lean/Tier33/Health/Transport/AttainedCapacity.lean::attainedHealthTransport_existsUnique_iff.`

Under Equation (26),

$$\boxed{Q_0(x) \bar{R} Q_1(y) \iff R(x, y).} \quad (27)$$

10

Determinacy constrains target observations across related pairs; saturation constrains the relation itself on both observation fibers.

5 Requirement refinement and information resolution

Let $L_0 \subseteq L_1$ be two requirement languages on one raw capacity carrier. Define

$$a \sim_{L_i} b \iff \forall r \in L_i, \text{Adequate}_r(a) \leftrightarrow \text{Adequate}_r(b). \quad (28)$$

Then

$$a \sim_{L_1} b \implies a \sim_{L_0} b,$$

so there is a canonical fine-to-coarse surjection

$$\boxed{\pi_{10} : \text{HealthCap}_{L_1} \twoheadrightarrow \text{HealthCap}_{L_0}, \quad [a]_{L_1} \longmapsto [a]_{L_0}.} \quad (29)$$

Theorem 5.1 (Reverse identity criterion). *An identity-induced commuting map*

$$\iota : \text{HealthCap}_{L_0} \rightarrow \text{HealthCap}_{L_1}, \quad \iota([a]_{L_0}) = [a]_{L_1},$$

exists exactly when

$$\boxed{\forall a, b, \quad a \sim_{L_0} b \iff a \sim_{L_1} b.} \quad (30)$$

Proof. The inclusion $L_0 \subseteq L_1$ already gives

$$a \sim_{L_1} b \implies a \sim_{L_0} b.$$

If ι exists and $a \sim_{L_0} b$, then

$$[a]_{L_0} = [b]_{L_0},$$

so

$$[a]_{L_1} = \iota([a]_{L_0}) = \iota([b]_{L_0}) = [b]_{L_1},$$

which gives $a \sim_{L_1} b$. The converse follows from Theorem 2.1. $\square$

11

Strict language growth can preserve the same information partition. In the finite control,

$$\text{Cap} = \text{Req} = \{\text{false}, \text{true}\}, \quad \text{Adequate}_r(c) \iff c = \text{true},$$

¹⁰Formal counterpart: `lean/Tier33/Health/Transport/RelationalHealth.lean::relationalHealthTransport_reflects_of_saturation.`

¹¹Formal counterpart: `lean/Tier33/Health/Paper/FAQ/ContextTransportObstruction.lean::reverseIdentityTransport_exists_iff.`

independently of r , with

$$L_0 = \{\text{false}\}, \quad L_1 = \{\text{false}, \text{true}\}.$$

The complete truth table is

	$r = \text{false}$	$r = \text{true}$
$c = \text{false}$	$\perp$	$\perp$
$c = \text{true}$	$\top$	$\top$

and therefore

$$\boxed{L_0 \subsetneq L_1, \quad \sim_{L_0} = \sim_{L_1} .} \quad (31)$$

Thus syntactic language growth can preserve the same information partition. ¹²

6 Explicit developmental contexts and stage–horizon separation

Use a distinct symbol for developmental context:

$$\boxed{\chi = (s, d, t), \quad \iota(\chi) = (\mathfrak{S}_s, d, t).} \quad (32)$$

A developmental family assigns

$$s \mapsto (X_s, \mathfrak{S}_s),$$

and its query requirements satisfy

$$r \in \mathcal{Q}(\mathfrak{S}_s).$$

A lawful developmental relation has the form

$$R_{s_0 s_1} : X_{s_0} \rightsquigarrow X_{s_1}. \quad (33)$$

¹³

The finite stage carrier is

$$\boxed{S_{\text{fix}} = \{N, A, E\} = \{\text{newborn}, \text{adult}, \text{elderly}\}.} \quad (34)$$

The declared stage law is exactly

$$\boxed{\text{StageLaw} = \{(N, A), (A, E)\}.} \quad (35)$$

These are exactly the lawful ordered pairs in the finite stage law.

The state carrier is

$$X = \{B, P, R, A_d, F\}$$

for baseline, perturbed, recovered, adapted, and failed. The capacity atoms are

$$\{\text{stable}, \text{exposed}\},$$

¹²Formal counterpart: `lean/Tier33/Health/Paper/FAQ/ContextTransportNegativeControls.lean::strict_language_growth_alone_does_not_obstruct_transport.`

¹³Formal counterpart: `lean/Tier33/Health/Development/Context.lean::DevelopmentalHealthContext;RelationalStageHealthTransport.`

and the requirements are

$$\{r_{\text{any}}, r_{\text{stable}}\}.$$

All three stages use the same underlying health specification:

$$\boxed{\mathfrak{S}_N = \mathfrak{S}_A = \mathfrak{S}_E.} \quad (36)$$

Hence, for every stage pair, scenario, horizon, and state,

$$\boxed{\text{Capacity}_{s_0,d,t}(x) = \text{Capacity}_{s_1,d,t}(x).} \quad (37)$$

14

The licensed fibers are, at every scenario and horizon,

$$\boxed{L_N = \{r_{\text{any}}\}, \quad L_A = L_E = \{r_{\text{any}}, r_{\text{stable}}\}.} \quad (38)$$

15

For each fixed stage and scenario,

$$\boxed{L_{s,d,\text{now}} = L_{s,d,\text{long}}, \quad \sim_{s,d,\text{now}} = \sim_{s,d,\text{long}}.} \quad (39)$$

16

The orthogonality square is therefore

$$\boxed{\begin{array}{ccc} (N, \text{mild}, \text{now}) & \xrightarrow{\text{stage}} & (A, \text{mild}, \text{now}) \\ \downarrow \text{horizon} & & \downarrow \text{horizon} \\ (N, \text{mild}, \text{long}) & \xrightarrow{\text{stage}} & (A, \text{mild}, \text{long}). \end{array}}$$

Along a horizontal arrow, raw capacity is unchanged and the licensed partition may change. Along a vertical arrow, the licensed partition is unchanged and raw capacity may change. For the perturbed state,

$$\text{Capacity}_{A,\text{mild},\text{now}}(P) = \{\text{exposed}\} \neq \{\text{stable}, \text{exposed}\} = \text{Capacity}_{A,\text{mild},\text{long}}(P). \quad (40)$$

17

In this fixture, the canonical raw-capacity map is generated by the now-to-long transition, and stage changes the health-visible partition. The finite stage fixture isolates stage-dependent health semantics with a fixed raw capacity law. General developmental families may carry stage-dependent states, specifications, dynamics, observers, and requirement types.

¹⁴Formal counterpart: `lean/Tier33/Health/Development/StageHorizonSeparation.lean::stage_specifications_equal;stage_change_preserves_raw_capacity.`

¹⁵Formal counterpart: `lean/Tier33/Health/Development/StageHorizonSeparation.lean::fixture_licensed_fiber.`

¹⁶Formal counterpart: `lean/Tier33/Health/Development/StageHorizonSeparation.lean::horizon_change_preserves_licensed_fiber;horizon_change_preserves_licensed_partition.`

¹⁷Formal counterpart: `lean/Tier33/Health/Development/StageHorizonSeparation.lean::stage_horizon_semantics_separate;canonical_capacity_map_is_horizon_generated.`

7 Complete finite capacity census and the canonical now-to-long map

Under the mild scenario define

$$C_0(x) = \text{Capacity}(x; \text{now}), \quad C_1(x) = \text{Capacity}(x; \text{long}).$$

The complete history semantics yield

x	$C_0(x)$	$C_1(x)$	$\text{Realizes}(x)$
B	$\{\text{stable}, \text{exposed}\}$	$\{\text{stable}, \text{exposed}\}$	$\top$
P	$\{\text{exposed}\}$	$\{\text{stable}, \text{exposed}\}$	$\top$
R	$\{\text{stable}\}$	$\{\text{stable}\}$	$\top$
A_d	$\{\text{stable}\}$	$\{\text{stable}\}$	$\top$
F	$\emptyset$	$\emptyset$	$\perp$

(41)

18

Every subset of the two-atom carrier occurs in the source column:

$$\emptyset, \quad \{\text{stable}\}, \quad \{\text{exposed}\}, \quad \{\text{stable}, \text{exposed}\}.$$

Thus

$$\boxed{C_0 : X \rightarrow \mathcal{P}\{\text{stable}, \text{exposed}\}.}$$
(42)

19

The complete induced map is therefore forced:

$$\boxed{\begin{aligned} \kappa_{\text{can}}(\emptyset) &= \emptyset, \\ \kappa_{\text{can}}(\{\text{stable}\}) &= \{\text{stable}\}, \\ \kappa_{\text{can}}(\{\text{exposed}\}) &= \{\text{stable}, \text{exposed}\}, \\ \kappa_{\text{can}}(\{\text{stable}, \text{exposed}\}) &= \{\text{stable}, \text{exposed}\}. \end{aligned}}$$
(43)

Equivalently,

$$\boxed{\kappa_{\text{can}}(c) = c \cup \{\text{stable} \mid c \neq \emptyset\}.}$$
(44)

The determining laws can be read directly from the census:

$$\text{stable} \in C_1(x) \iff C_0(x) \neq \emptyset, \tag{45}$$

$$\text{exposed} \in C_1(x) \iff \text{exposed} \in C_0(x). \tag{46}$$

Hence $C_0(x) = C_0(x')$ implies equality of membership for both target atoms, so $C_1(x) = C_1(x')$. Capacity determinacy follows.

¹⁸Formal counterpart: `lean/Tier33/Health/Development/CanonicalRecovery.lean::sourceCapacity_inventory;targetCapacity_inventory.`

¹⁹Formal counterpart:
`lean/Tier33/Health/Development/CanonicalRecovery.lean::sourceCapacity_surjective.`

Theorem 7.1 (Global canonicity). *There exists a unique*

$$\kappa : \mathcal{P}\{\text{stable}, \text{exposed}\} \rightarrow \mathcal{P}\{\text{stable}, \text{exposed}\}$$

such that

$$\kappa(C_0(x)) = C_1(x) \quad \forall x \in X.$$

It is the map in Equations (43)–(44).

Proof. Theorem 3.2 constructs the unique map on relation-attained source capacities. Here the state relation is graph identity,

$$R(x, y) \iff y = x,$$

so every source state participates in a related pair. Equation (42) therefore implies

$$\mathcal{A}_R = \mathcal{A} = \mathcal{P}\{\text{stable}, \text{exposed}\}.$$

Equation (20) promotes attained uniqueness to global uniqueness. Each row of Equation (41) fixes one of the four possible input values, giving Equation (43). $\square$

20 21

The previously defined operation

$$\text{recoverCapacity}(c) = c \cup \{\text{stable} \mid c \neq \emptyset\}$$

therefore satisfies

$$\boxed{\text{recoverCapacity} = \kappa_{\text{can}}.} \tag{47}$$

It is the unique global capacity map commuting with all complete-history source/target capacities in this fixture. ²²

The canonicity claim is representation-level:

$$\boxed{\begin{array}{l} \kappa_{\text{can}} \text{ is unique on the declared capacity carrier,} \\ \text{and its canonicity is confined to the declared capacity representation.} \end{array}} \tag{48}$$

The complete viable-history family remains the upstream object defined in Paper I; the present uniqueness theorem concerns only the downstream capacity representation induced from it.

²⁰Formal counterpart: `lean/Tier33/Health/Transport/AttainedCapacity.lean::capacityMap_unique_of_sourceCapacity_surjective.`

²¹Formal counterpart: `lean/Tier33/Health/Development/CanonicalRecovery.lean::canonicalTransport_unique;recoverCapacity_unique_global.`

²²Formal counterpart: `lean/Tier33/Health/Development/CanonicalRecovery.lean::recoverCapacity_eq_canonicalTransport;recoverCapacity_unique_of_capacityCommutes.`

8 Stage-only class splitting

At mild/now, the perturbed and recovered capacities are

$$C(P) = \{\text{exposed}\}, \quad C(R) = \{\text{stable}\}.$$

The exact adequacy table is

	newborn r_{any}	adult r_{any}	adult r_{stable}
$C(P)$	$\top$	$\top$	$\perp$
$C(R)$	$\top$	$\top$	$\top$

(49)

because r_{any} tests nonemptiness and r_{stable} tests membership of the stable atom.

Thus

$$\boxed{Q_N(P) = Q_N(R), \quad Q_A(P) \neq Q_A(R).} \quad (50)$$

The raw capacity itself is unchanged by the stage change, by Equation (37).

Theorem 8.1 (Stage-only attained obstruction). *At fixed mild/now, graph identity from newborn to adult has an attained health-class obstruction to deterministic transport.*

Proof. The two related pairs

$$(P, P), \quad (R, R)$$

have the same newborn source class by Equation (50) and different adult target classes. Equation (19), applied to Q_N, Q_A , gives the obstruction. $\square$

²³

The adult-to-elderly transition is lawful and

$$L_A = L_E, \quad \sim_A = \sim_E,$$

which exhibits a lawful stage change with preserved health-visible information. ²⁴

9 Branching, coalescence, and the value of the health quotient

9.1 Branching

At adult/mild/now define

$$\boxed{R_{\text{branch}} = \{(P, P), (P, R)\}.} \quad (51)$$

Equation (49) gives

$$Q_A(P) \neq Q_A(R).$$

²³Formal counterpart: `lean/Tier33/Health/Development/RelationalControls.lean::stage_only_source_classes_merge;stage_only_healthTransport_obstructed.`

²⁴Formal counterpart: `lean/Tier33/Health/Development/StageFixture.lean::neutral_stage_change_need_not_change_health_information.`

The relation-valued image contains both

$$Q_A(P) \bar{R}_{\text{branch}} Q_A(P), \quad Q_A(P) \bar{R}_{\text{branch}} Q_A(R). \quad (52)$$

Hence $\bar{R}_{\text{branch}}$ has two target classes at the attained source class $Q_A(P)$:

$$\boxed{\neg \text{HealthClassDeterminate}(R_{\text{branch}}; Q_A, Q_A).} \quad (53)$$

The relation-valued transport remains defined and retains both edges. ²⁵

9.2 Coalescence

Define

$$\boxed{R_{\text{coal}} = \{(x, R) \mid x \in X\}.} \quad (54)$$

For example,

$$Q_A(P) \bar{R}_{\text{coal}} Q_A(R), \quad Q_A(R) \bar{R}_{\text{coal}} Q_A(R).$$

Every related target is R , so the image has one target class for each source class. Therefore forward transport is functional. ²⁶

Branching and coalescence therefore have different representational consequences:

$$1 \rightarrow n \quad \text{may destroy functionality,} \quad n \rightarrow 1 \quad \text{may preserve it.}$$

9.3 Raw indeterminacy with health-visible determinacy

Define

$$\boxed{R_{\text{raw}} = \{(P, B), (P, R)\}.} \quad (55)$$

The target raw capacities are

$$C(B) = \{\text{stable}, \text{exposed}\}, \quad C(R) = \{\text{stable}\},$$

so

$$C(B) \neq C(R).$$

Thus raw capacity determinacy fails.

The adult truth table is

	r_{any}	r_{stable}
$C(B) = \{\text{stable}, \text{exposed}\}$	$\top$	$\top$
$C(R) = \{\text{stable}\}$	$\top$	$\top$

(56)

so

$$Q_A(B) = Q_A(R).$$

²⁵Formal counterpart: `lean/Tier33/Health/Development/RelationalControls.lean::branching_relation_valued_transport_survives;branching_health_classes_obstruct_function.`

²⁶Formal counterpart: `lean/Tier33/Health/Development/RelationalControls.lean::coalescence_can_preserve_forward_determinism.`

Hence every target of R_{raw} has the same health-visible class, and

$$\boxed{\neg \text{CapacityDeterminate}(R_{\text{raw}}, C, C), \quad \text{HealthClassDeterminate}(R_{\text{raw}}; Q_A, Q_A).} \quad (57)$$

27

Equation (57) gives the central role of the health quotient: the raw dynamical distinction remains nondeterministic, and the distinctions consumed by all licensed health queries transport deterministically.

10 Combined stage–horizon transport

Consider

$$\chi_0 = (N, \text{mild}, \text{now}), \quad \chi_1 = (A, \text{mild}, \text{long}),$$

with stage law $N \rightarrow A$ and state relation

$$\boxed{R_{NA}(x, y) \iff y = x.} \quad (58)$$

Theorem 10.1 (Raw determinacy for the combined transition). *For any x, x' ,*

$$C_0(x) = C_0(x') \implies C_1(x) = C_1(x').$$

Consequently there exists a unique

$$\kappa_{NA} : \text{Attained}_{R_{NA}}(C_0) \rightarrow \text{Cap}_{A, \text{long}}$$

with

$$\kappa_{NA}(C_0(x)) = C_1(x).$$

Proof. Equality of source capacities implies equal nonemptiness and equal exposed membership. Equations (45)–(46) therefore give equality of the two target atom-membership predicates, hence equality of target capacities. Theorem 3.2 supplies the unique attained factorization. $\square$

28

Theorem 10.2 (Health-class determinacy for the combined transition). *If*

$$Q_N(x) = Q_N(x'),$$

then

$$Q_{A, \text{long}}(x) = Q_{A, \text{long}}(x').$$

Consequently there exists a unique attained health-visible transport

$$T_{NA} : \text{Attained}_{R_{NA}}(Q_N) \rightarrow \text{HealthCap}_{A, \text{long}}.$$

²⁷ Formal counterpart:

```
lean/Tier33/Health/Development/RelationalControls.lean::raw_branching_obstructs_capacity_transport
;baseline_recovered_health_classes_equal;raw_indeterminacy_can_be_health_determinate.
```

²⁸ Formal counterpart:

```
lean/Tier33/Health/Development/StageFixture.lean::combined_capacityTransport_existsUnique.
```

Proof. Newborn governance contains only r_{any} , so

$$Q_N(x) = Q_N(x')$$

means precisely that $C_0(x)$ and $C_0(x')$ have equal nonemptiness truth. At adult/long, both licensed requirements have that same source truth:

$$r_{\text{any}} \iff C_0(x) \neq \emptyset,$$

and by Equation (45),

$$r_{\text{stable}} \iff C_0(x) \neq \emptyset.$$

Therefore all adult target licensed predicates agree. Theorem 3.2, applied to the Q -observations, gives the unique attained map. $\square$

29

Raw and health-visible determinacy are established as separate criteria.

11 From health-visible transport to full health truth

Relation-valued health-class transport records prospective capacity information. Full health additionally contains constitutive realization.

Let

$$R : X_0 \rightsquigarrow X_1$$

and

$$\rho : \mathcal{Q}(\mathfrak{S}_{c_0}) \rightarrow \mathcal{Q}(\mathfrak{S}_{c_1}).$$

Assume licensing coherence

$$r \in L_{c_0} \implies \rho(r) \in L_{c_1}, \quad (59)$$

realization compatibility

$$R(x, y) \implies (\text{Realizes}_{c_0}(x) \leftrightarrow \text{Realizes}_{c_1}(y)), \quad (60)$$

and adequacy compatibility

$$R(x, y) \implies (\overline{\text{Adequate}}_{c_0, r}(Q_0(x)) \leftrightarrow \overline{\text{Adequate}}_{c_1, \rho(r)}(Q_1(y))) \quad (61)$$

for every source licensed requirement r .

Proposition 11.1 (Full relational health truth). *For every $R(x, y)$ and $r \in L_{c_0}$,*

$$\boxed{\text{Health}_{c_0, r}(x) \iff \text{Health}_{c_1, \rho(r)}(y)}. \quad (62)$$

²⁹ Formal counterpart: `lean/Tier33/Health/Development/StageFixture.lean::combined_healthClassDeterminate;combined_healthTransport_existsUnique.`

Proof. Using Equation (9),

$$\begin{aligned}
\text{Health}_{c_0, r}(x) &\iff \text{Realizes}_{c_0}(x) \wedge \overline{\text{Adequate}}_{c_0, r}(Q_0(x)) \\
&\iff \text{Realizes}_{c_1}(y) \wedge \overline{\text{Adequate}}_{c_1, \rho(r)}(Q_1(y)) \\
&\iff \text{Health}_{c_1, \rho(r)}(y).
\end{aligned}$$

The middle equivalence uses Equations (60) and (61). $\square$

30

Proposition 11.1 is an interface corollary of the two clauses in the immutable health normal form and is stated directly for related states. Realization and adequacy compatibility establish the propositional truth equivalence. Licensing coherence plays a governance and typing role: it guarantees that the translated requirement $\rho(r)$ remains inside the target licensed query family, so the descended target adequacy predicate is interpreted within the governed language.

12 Open-system support: raw, visible, and full-health controls

The support state is

$$z = (o, s), \quad o \in \{\text{false}, \text{true}\},$$

with

$$s \in \{\text{endogenous}, \text{persistent}, \text{intermittent}, \text{withdrawn}\}.$$

The route carrier is

$$\{\text{reserve}, \text{autonomous}\}.$$

The two adequacy predicates are

$$\boxed{\text{Adequate}_{\text{reserve}}(c) \iff \text{reserve} \in c, \quad \text{Adequate}_{\text{autonomous}}(c) \iff \text{autonomous} \in c.} \quad (63)$$

All route requirements are licensed at both horizons.

The complete capacity table is

z	$C_{\text{short}}(z)$	$C_{\text{long}}(z)$	$\text{Realizes}(z)$
$(\top, \text{endogenous})$	$\{\text{reserve}, \text{autonomous}\}$	$\{\text{reserve}, \text{autonomous}\}$	$\top$
$z_P = (\top, \text{persistent})$	$\{\text{reserve}\}$	$\{\text{reserve}\}$	$\top$
$z_I = (\top, \text{intermittent})$	$\{\text{reserve}\}$	$\emptyset$	$\top$
$(\top, \text{withdrawn})$	$\emptyset$	$\emptyset$	$\top$
$(\perp, s)$	$\emptyset$	$\emptyset$	$\perp$

(64)

For graph identity across horizon,

$$C_{\text{short}}(z_P) = C_{\text{short}}(z_I) = \{\text{reserve}\},$$

³⁰ Formal counterpart:

`lean/Tier33/Health/Transport/RelationalHealthTruth.lean::related_health_truth_equiv.`

and

$$C_{\text{long}}(z_P) = \{\text{reserve}\} \neq \emptyset = C_{\text{long}}(z_I).$$

Therefore

$$\boxed{\neg \text{CapacityDeterminate}(R_{\text{horizon}}, C_{\text{short}}, C_{\text{long}}).} \quad (65)$$

31

At the quotient level,

$$Q_{\text{short}}(z_P) = Q_{\text{short}}(z_I),$$

and reserve adequacy separates the long-horizon classes:

$$Q_{\text{long}}(z_P) \neq Q_{\text{long}}(z_I).$$

Hence

$$\boxed{\neg \text{HealthClassDeterminate}(R_{\text{horizon}}; Q_{\text{short}}, Q_{\text{long}}).} \quad (66)$$

32

12.1 Withdrawal

Define

$$\text{withdraw}(o, s) = (o, \text{withdrawn}), \quad R_{\text{wd}} = \text{graph}(\text{withdraw}),$$

and

$$\boxed{\kappa_{\text{wd}}(a) = \emptyset.} \quad (67)$$

For every state,

$$\boxed{\kappa_{\text{wd}}(\text{Capacity}(z)) = \text{Capacity}(\text{withdraw}(z)) = \emptyset.} \quad (68)$$

Thus raw quotient descent exists and the attained health-class image is functional.

For $z_P = (\top, \text{persistent})$ at long horizon,

state	Realizes	$r_{\text{reserve}} \in L$	Adequate _{reserve}	Health _{reserve}
z_P	$\top$	$\top$	$\top$	$\top$
$\text{withdraw}(z_P)$	$\top$	$\top$	$\perp$	$\perp$

The relation preserves realization and licensing. Adequacy compatibility would require

$$\text{reserve} \in \{\text{reserve}\} \leftrightarrow \text{reserve} \in \emptyset,$$

which is

$$\top \leftrightarrow \perp.$$

³¹Formal counterpart: `lean/Tier33/Health/Transport/SupportFactorization.lean::support_horizon_obstruction_is_capacityFactorization_obstruction.`

³²Formal counterpart: `lean/Tier33/Health/Transport/SupportFactorization.lean::support_horizon_obstruction_is_healthFactorization_obstruction.`

Therefore

$$\boxed{\begin{array}{c} \text{functional health-class transport} + \text{realization compatibility} + \text{licensing coherence} \\ \not\Rightarrow \text{full health preservation} \\ \text{when adequacy compatibility fails.} \end{array}} \quad (69)$$

33

12.2 Governed query-family aggregates

The contextual transport theory operates on individual governed queries. Quantified aggregates over query families remain available through

$$\text{UniformHealth}_K(x) := \forall q \in K, \text{Health}_q(x), \quad \text{ExistentialHealth}_K(x) := \exists q \in K, \text{Health}_q(x),$$

with neighborhood variants defined by restriction to a declared query relation. These are ordinary quantifier constructions; metric, probabilistic, perturbational, and adversarial robustness belong to separate semantics. Their elementary logical properties and finite countermodels are collected in Appendix A.

13 Related work and contribution

13.1 Health, capability, and context

Action-theoretic accounts characterize health through ability relative to vital goals and circumstances [4, 5]. Biostatistical accounts center statistically normal biological functioning [3]. Capability approaches connect health to agency, achievable functionings, and surrounding circumstances [6, 7]. The present formalism treats scientific selection of requirements as a governance input and studies the information structure induced by that selection.

The WHO International Classification of Functioning, Disability and Health places functioning within an interaction among health conditions, personal factors, and environmental factors [8]. The support model here addresses a formal representation question: whether support-dependent prospective capacity determines a target raw capacity, a health-visible class, and finally a health proposition under contextual change.

13.2 Development and relation-valued change

Developmental biology is stage-structured and historically organized [9]. The present formalism separates stage, horizon, state relation, capacity observation, governance, and health truth. Relation-valued stage change admits branching and coalescence. Deterministic factorization is established by an additional theorem.

³³*Formal counterpart:* `lean/Tier33/Health/Transport/SupportHealthTruth.lean::withdrawal_healthClass_determine;withdrawal_adequacy_incompatible;healthClass_transport_alone_does_not_imply_healthTruth.`

13.3 Institutions, abstraction, and observational equivalence

Institution theory provides an abstract setting for signatures, translations, and satisfaction invariance [10]. Predicate abstraction and observational equivalence provide established mathematical precedents for quotienting by a family of observations [11, 12]. The quotient and factorization principles used here are standard. The health-specific construction is

$$\begin{array}{l}
 \text{whole-query licensing} \longrightarrow \text{context/stage-specific requirement fibers} \\
 \longrightarrow \text{predicate-generated health equivalence} \longrightarrow \text{health-visible quotients} \\
 \longrightarrow \text{comparison with relation-generated capacity evolution} \\
 \longrightarrow \text{exact preservation or loss of health-relevant information.}
 \end{array} \tag{70}$$

The finite models expose the semantic layers directly:

$$\begin{array}{l}
 \text{stage and horizon act through distinct contextual coordinates,} \\
 \text{raw and health-class determinacy test distinct observations,} \\
 \text{relation-valued and functional transport occupy distinct levels,} \\
 \text{full health adds realization and adequacy compatibility.}
 \end{array} \tag{71}$$

These theorem-level distinctions locate the semantic layers of contextual transport.

14 Discussion

The central structural result is the separation of transport questions by representation level. Raw capacity transport is controlled by

$$\text{CapacityDeterminate}(R, C_0, C_1),$$

health-visible transport by

$$\text{HealthClassDeterminate}(R; Q_0, Q_1),$$

and full health comparison by realization compatibility together with licensed adequacy compatibility. These criteria answer distinct scientific questions.

Equation (57) gives the clearest health-specific consequence. Baseline and recovered states remain distinguishable at the raw capacity level and share the same adult licensed adequacy signature. The quotient retains exactly the distinctions consumed by the licensed health language, allowing deterministic transport of health-visible information across a relation with raw-capacity indeterminacy.

The finite developmental square provides a controlled decomposition of contextual effects. Stage changes the licensed evaluation language under a fixed raw capacity law; horizon changes the generated prospective capacity under a fixed licensed partition. Their combined transition carries both mechanisms. The canonical now-to-long map is unique on the declared capacity representation because graph identity supplies relation-attained coverage and the source capacity census is surjective.

The support model carries the same factorization logic into an open-system setting. Persistent and intermittent support coincide at short horizon and separate at long horizon. Withdrawal preserves

constitutive realization and the licensed requirement family and changes reserve adequacy. The full-health theorem identifies adequacy compatibility as the additional semantic condition required to transport the health proposition itself.

15 Scope boundaries

The theorem prover verifies consequences relative to declared dynamics, constitutive semantics, query governance, stage relations, support semantics, and adequacy predicates. Biological and clinical correctness remain independent scientific obligations.

The finite developmental stages N, A, E are formal stage labels. The stage/horizon separation, canonical capacity census, branching controls, and coalescence controls are formal models whose empirical interpretation requires separate evidence.

The following structures remain outside Paper II:

endogenous control and admissible policy,
resource and history cost,
route redundancy and independence structure,
probability-sensitive capacity and risk,
failure-aware and perturbational robustness semantics.

Measurement identification, empirical query validation, calibration, uncertainty, and predictive comparison form the later measurement layer.

16 Conclusion

Paper II extends the fixed-query architecture of Paper I to changing contexts. Contextual licensing generates

$$\text{HealthCap}_c = \text{Cap}_c / \sim_c,$$

state relations generate attained transport problems, and determinacy characterizes unique maps at the raw and health-visible levels. The developmental and support fixtures make factorization, obstruction, branching, and full-health compatibility explicit.

The resulting hierarchy is

state relation $\longrightarrow$ capacity transport $\longrightarrow$ health-visible transport $\longrightarrow$ full health comparison.
--

Lean certifies the displayed argument and finite witnesses.

A Governed query-family aggregates

For a governed query family K ,

$$\text{UniformHealth}_K(x) := \forall q \in K, \text{Health}_q(x), \quad \text{ExistentialHealth}_K(x) := \exists q \in K, \text{Health}_q(x).$$

For a neighborhood relation N ,

$$\text{NeighborhoodHealth}_{K,N}(q_0, x) := q_0 \in K \wedge \forall q \in K, N(q_0, q) \rightarrow \text{Health}_q(x).$$

Uniform health implies existential health for nonempty K . Empty-family uniform truth is vacuous and empty-family existential truth is false. Neighborhood health implies the anchor query when the neighborhood relation is reflexive at the anchor. These are ordinary predicate-logic consequences.

In the finite aggregate fixture, stable-reserve health at the perturbed state is false for mild/now and true for mild/long. Selected-long and existential truth therefore hold, and uniform truth over the two-query family fails. This gives a strict finite separation among the aggregate notions within ordinary quantifier semantics. ³⁴

B Structural developmental-stage interoperability

A separate developmental-stage calculus uses

$$S_{\text{RMMO}} = \{E9.5, E10.5, E11.5, E12.5, E13.5, E14.5, E15.5, E16.5\}$$

with adjacency

$$s \prec s' \iff \text{rank}(s') = \text{rank}(s) + 1.$$

Suppose independent health specifications $\mathfrak{S}_s$ are supplied on the same stage-indexed configuration carrier. The structural adapter preserves the original relation literally:

$$\boxed{R_{s,s'}^{\text{Health}} = R_{s,s'}^{\text{RMMO}}.} \quad (72)$$

³⁵

Equation (72) supplies the structural relation identity. Licensed requirement families, adequacy predicates, determinacy results, health preservation, lineage interpretation, and empirical validation remain independent inputs.

C Finite-history rules underlying the developmental census

The finite state carrier is

$$X = \{B, P, R, A_d, F\}.$$

³⁴Formal counterpart: `lean/Tier33/Health/Paper/FAQ/HigherOrderHealth.lean::UniformHealth;queryRobustHealth_eq_uniformHealth;universal_implies_existential;universal_empty;existential_empty.`

³⁵Formal counterpart: `lean/Tier33/Health/Bridge/RelationalMorphogenesisStage.lean::registered_relation_preserved;family_relation_preserved.`

At every scenario/horizon each state has a stay history. Under mild conditions, baseline has a perturbing history $B \rightarrow P$. Under mild/long, P has recovery and adaptation histories

$$P \rightarrow R, \quad P \rightarrow A_d.$$

Under severe/long, P has a lawful failure history

$$P \rightarrow F.$$

Continuation requires the endpoint to realize the declared organization. Histories ending in F therefore lie outside viable capacity. The observer assigns:

$$P \mapsto \text{exposed}, \quad B, R, A_d \mapsto \text{stable},$$

with the perturbing history observed as exposed and recovery/adaptation histories observed as stable. Exhausting these constructors yields Equation (41).

D Formal authority and verification

The preferred Paper-II entry point is

`lean/Tier33/Health/PaperII.lean.`

The stage, transport, and full-health theorem surface is pinned at commit

`cfc6eb4b5b070537b16f790d307e5dddfcd072e9.`

The final archival DOI and public source/reproduction location will be added to the publication version together with the immutable release tag; the commit above remains the exact formal-source pin. The checked environment is

`Lean 4.28.0-rc1, mathlib 5352afccd6866369be9de43f5b7ec47203555f44.`

One-command verification is

`python3 scripts/verify_t33_health_relhealth03.py -compile.`

It checks the imported-source boundary, Paper-I preservation, the preferred Paper-II dependency cut, theorem ownership, kernel inventories, and elaboration of the planned manuscript references.

The RELHEALTH03 gate executes 15 explicit Lean invocations. The new relational-health audit contains 20 declarations; the developmental audit contains 68; the contextual transport audit contains 81; and the integrated FAQ audit contains 88. The allowed kernel dependency inventory is

`{propext, Quot.sound, Classical.choice}.`

The verification surface uses the original `Health.Health` definition and Paper-I source together with kernel-checked proofs free of project axioms, proof admissions, compiler-trust shortcuts, proof-via-`native_decide`, and caller-selected favorable transport maps.

The repository configuration leaves the default Lake target empty, so literal `lake build` performs a zero-job environment check. Proof coverage comes from the explicit scoped invocations above.

Scope. Machine checking certifies the displayed formal consequences relative to the declared semantics. Scientific licensing, biological interpretation, physiological measurement validity, and empirical validation require independent evidence.

E Reader theorem map

ID	Section	Mathematical statement	Formal counterpart
PHC-01	Quotient descent	Compatibility iff unique commuting contextual quotient map.	<code>ContextTransport.lean::contextQuotientTransport_existsUnique_iff</code>
PHC-02	Quotient descent	Split source fibers exactly obstruct commuting quotient descent.	<code>ContextTransportObstruction.lean::contextQuotientTransport_obstruction_iff_splitFiber</code>
PHC-03	Attained factorization	Capacity determinacy iff unique relation-attained factorization.	<code>AttainedCapacity.lean::attainedCapacityTransport_existsUnique_iff</code>
PHC-04	Attained factorization	Equal source observation with unequal target observation exactly obstructs deterministic factorization.	<code>AttainedCapacity.lean::attainedCapacityTransport_obstruction_iff</code>
PHC-05	Relational transport	The health-class image relation is left-total on attained source classes.	<code>RelationalHealth.lean::attainedRelationalHealthTransport_leftTotal</code>
PHC-06	Relational transport	Functional attained health-class transport iff health-class determinacy.	<code>RelationalHealth.lean::relationalHealthTransport_functional_iff_determinate</code>
PHC-07	Refinement	Reverse identity transport exists iff coarse and fine languages induce the same equivalence relation.	<code>ContextTransportObstruction.lean::reverseIdentityTransport_exists_iff</code>
PHC-08	Stage/horizon	Stage changes preserve raw capacity in the finite fixture and can change stage governance.	<code>StageHorizonSeparation.lean::stage_change_preserves_raw_capacity;stage_can_change_licensed_fiber</code>
PHC-09	Stage/horizon	Horizon changes preserve the licensed partition and can change perturbed raw capacity.	<code>StageHorizonSeparation.lean::horizon_change_preserves_licensed_partition;horizon_only_capacity_can_change</code>
PHC-10	Canonical map	The current-horizon capacity function is surjective over the complete two-atom carrier.	<code>CanonicalRecovery.lean::sourceCapacity_surjective</code>
PHC-11	Canonical map	Complete source/target capacities determine a unique global map equal to recoverCapacity.	<code>CanonicalRecovery.lean::recoverCapacity_unique_global;recoverCapacity_eq_canonicalTransport</code>
PHC-12	Stage-only split	Newborn equivalence merges P/R; adult stable reserve separates them.	<code>RelationalControls.lean::stage_only_healthTransport_obstructed</code>
PHC-13	Branching	Relation-valued transport survives a branch that obstructs functional transport.	<code>RelationalControls.lean::branching_relation_valued_transport_survives;branching_health_classes_obstruct_function</code>
PHC-14	Raw/visible separation	Raw indeterminacy can coexist with health-class determinacy.	<code>RelationalControls.lean::raw_indeterminacy_can_be_health_determinate</code>

ID	Section	Mathematical statement	Formal counterpart
PHC-15	Combined transition	Newborn/now to adult/long admits unique attained raw and health-visible transports.	<code>StageFixture.lean::combined_capacityTransport_existsUnique; combined_healthTransport_existsUnique</code>
PHC-16	Full health	Relation-level realization and licensed adequacy compatibility imply full health-truth equivalence.	<code>RelationalHealthTruth.lean::related_health_truth_equiv</code>
PHC-17	Support horizon	Persistent/intermittent equality at short horizon and separation at long obstruct both raw and health-visible factorization.	<code>SupportFactorization.lean::support_horizon_obstruction_is_healthFactorization_obstruction</code>
PHC-18	Withdrawal	Functional health-class transport with realization/licensing can still fail full health when adequacy compatibility fails.	<code>SupportHealthTruth.lean::healthClass_transport_alone_does_not_imply_healthTruth</code>

F Theorem-classification guide

Class	Paper-II examples
Definition / interface	Contextual licensed fiber, developmental context, relation-valued health-class image, relation-level realization/licensing/adequacy compatibility.
Standard generic mathematics	Quotient descent, attained observation factorization, functional iff left-total and single-valued.
Health-specific representation theorem	Context-indexed quotient transport, attained health-class transport, stage-indexed visible transport.
Obstruction theorem	Split quotient fibers, relation-attained source collapse with target split, stage-only visible split, support horizon split.
Finite canonicity theorem	Complete capacity census, source surjectivity, unique global now-to-long map, equality with recoverCapacity.
Finite relational control	Explicit branching, coalescence, and raw-indeterminate/health-determinate relations.
Interface corollary	Full health-truth equivalence from realization and licensed adequacy compatibility.
External scientific obligation	Licensing criteria, biological stage interpretation, physiological measurement sufficiency, and empirical validation.

References

- [1] Z. James. Constitutive Continuation Capacity: A machine-checked framework for prospective health. Preprint, 2026. doi:10.5281/zenodo.23131158.
- [2] M. Huber, J. A. Knottnerus, L. Green, H. van der Horst, A. R. Jadad, D. Kromhout, B. Leonard, K. Lorig, M. I. Loureiro, J. W. M. van der Meer, P. Schnabel, R. Smith, C. van Weel, and H. Smid. How should we define health? *BMJ*, 343:d4163, 2011. doi:10.1136/bmj.d4163.
- [3] C. Boorse. Health as a theoretical concept. *Philosophy of Science*, 44(4):542–573, 1977. doi:10.1086/288768.
- [4] L. Nordenfelt. *On the Nature of Health: An Action-Theoretic Approach*. 2nd edition, Springer, Dordrecht, 1995. doi:10.1007/978-94-011-0241-4.
- [5] L. Nordenfelt. On the notion of health as ability. *Scandinavian Journal of Occupational Therapy*, 3:101–105, 1996. Reprinted in volume 21(sup1):48–52, 2014, doi:10.3109/11038128.2014.952902.
- [6] J. P. Ruger. Health capability: Conceptualization and operationalization. *American Journal of Public Health*, 100(1):41–49, 2010. doi:10.2105/AJPH.2008.143651.
- [7] S. Venkatapuram. Health, vital goals, and central human capabilities. *Bioethics*, 27(5):271–279, 2013. doi:10.1111/j.1467-8519.2011.01953.x.
- [8] World Health Organization. *International Classification of Functioning, Disability and Health: ICF*. World Health Organization, Geneva, 2001.
- [9] C. H. Waddington. *The Strategy of the Genes*. George Allen & Unwin, London, 1957.
- [10] J. A. Goguen and R. M. Burstall. Institutions: Abstract model theory for specification and programming. *Journal of the ACM*, 39(1):95–146, 1992. doi:10.1145/147508.147524.
- [11] P. Cousot and R. Cousot. Abstract interpretation: A unified lattice model for static analysis of programs by construction or approximation of fixpoints. In *Proceedings of the 4th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*, pages 238–252, 1977. doi:10.1145/512950.512973.
- [12] M. Hennessy and R. Milner. Algebraic laws for nondeterminism and concurrency. *Journal of the ACM*, 32(1):137–161, 1985. doi:10.1145/2455.2460.
- [13] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, Lecture Notes in Computer Science **12699**, Springer (2021) 625–635.
- [14] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, ACM (2020) 367–381, [arXiv:1910.09336](https://arxiv.org/abs/1910.09336) [cs.LO].